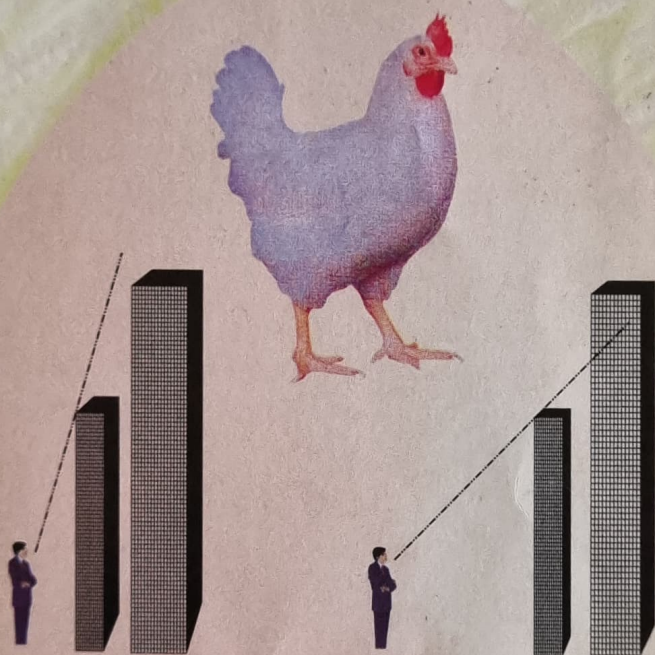


ENDOMETH

Look beyond synthetic methionine

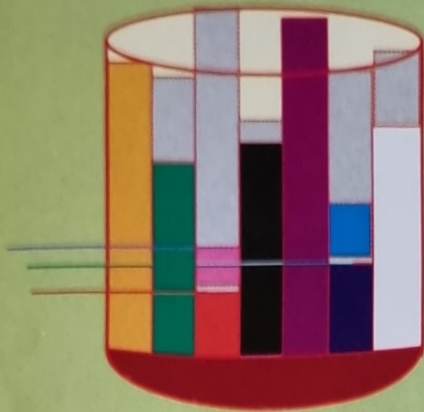


*Take a few steps back, look again and you will
find a much bigger and better picture.*

Look at Enterokinase & Methionine-Reconversion

Sometimes, in our enthusiasm we give too much importance to certain aspects meanwhile missing out on the others. In our enthusiasm over the beneficial effects of methionine, we are concentrating too much on it. However, we have failed to take in to notice the possibilities of making ENTEROKINASE and METHIONINE RECONVERSION work for our benefits.

Benefits of Enterokinase Stimulation



Methionine being the first limiting amino acid in poultry diets, all the other amino acids present in the dietary protein will be utilized only to the extent of the available methionine. Remaining amino acids will be deaminated and used for energy process, which is relatively expensive.

To avoid this wasteful protein metabolism and to turn it into a benefit there are two ways. One, increase the available methionine content by supplementing crystalline methionine. Two, improve the overall availability of natural dietary methionine by an integrated approach.

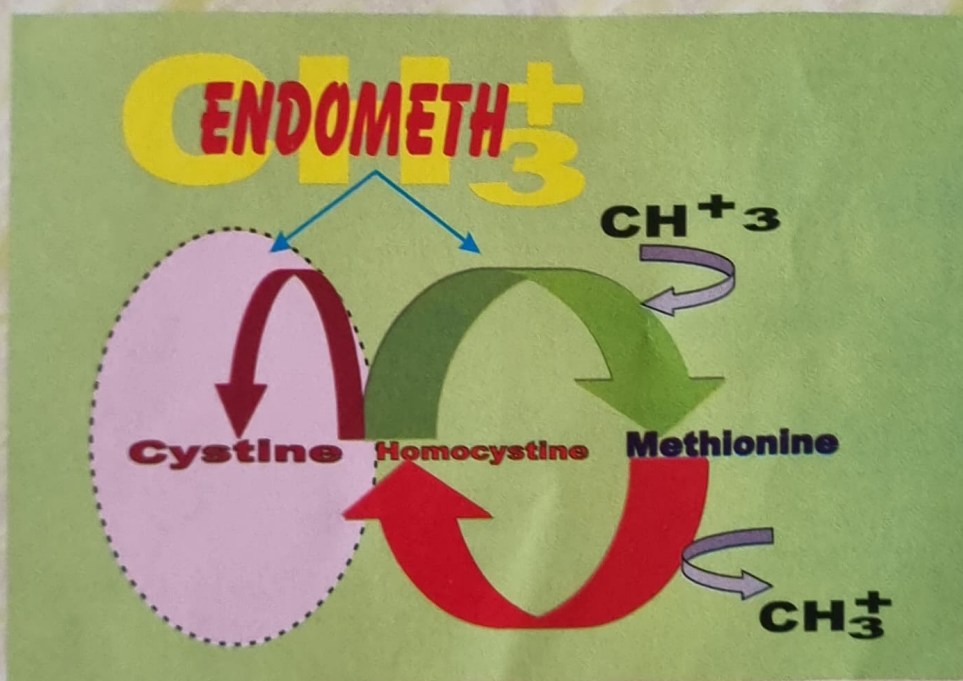
Approach 1 - Supplementing Synthetic Methionine: While supplementing synthetic methionine, we are considering that a calculated proportion of methionine is coming from the feed ingredients. However, due to variation in the quality of ingredients, most of the times we may not get the amount of methionine we expected from the dietary source, even though it may be physically present. Thus, even though we are supplementing a high level of synthetic methionine, there might be still a deficit of methionine, which brings down the performance of the birds greatly.

Approach 2 -Maximizing Methionine Utilization: Supplying protease, though, could help to certain extent the benefit is only marginal. Eliciting a greater enterokinase response to the presence of food thereby improving the native protease activity is a possibility which appears to be significantly beneficial. In addition, maximizing the conversion of trypsinogen to trypsin helps in two ways (1) better protein digestibility leading to reduced requirement for methionine supply and (2) Reduced requirement for Trypsinogen production (Trypsinogen has a very high sulphur amino acid content and requires higher amounts of methionine for its synthesis).

In both the above situations, **Vesper's ENDOMETH** helps the bird's system to spare valuable methionine in poultry diets. Used in replacement of the crystalline amino acid methionine, it can bring about better utilization of methionine, protein and hence all the nutrients for a better productivity and FCR.

Enhanced Methionine Reconversion

Whenever the body requires methyl groups, methionine is stripped of a methyl group and in the process it is converted into Homocystine. This homocystine can either be converted into cystine, a not-so-essential amino acid as compared to methionine, or it can be reconverted into methionine in the presence of a labile methyl donor donating methyl group. Many a times, even in the presence of methyl groups, the reconversion of homocystine to methionine may not take place due to lack of activation of the proper enzyme system involved. Aminonitrogen in Endometh activates the corresponding enzyme system and thereby promotes the reconversion of homocystine into methionine. Thus, Endometh can reduce the requirement of Synthetic methionine in poultry diets. Used in total replacement of the crystalline amino acid methionine, it can bring about better utilization of methionine, protein and hence all the nutrients for a better productivity and FCR.



Note: While you are using supplemental DL-methionine, you are not doing anything to improve the reconversion of methionine (both natural and synthetic). Thus, the total methionine content of your diet is not being used its fullest capacity. Endometh helps you maximize the reconversion of methionine from the feed. Therefore, you can effectively exploit the natural methionine activity for better performance and economy.

Brief Technical Information: Contents: Methyl donors and Aminonitrogen in a base enriched with Natural enhancers of Enterokinase Activity,.

Uses: To enhance the methionine value of feedstuffs. Usage Instructions: In synthetic methionine supplemented diets - Replace total synthetic methionine content with equal quantity of Endometh. In non - supplemented diets - Add 100-150gm of Endometh for better feed utilization, productivity and FCR.

Availability: 20 Kg bags.

Trial on the Efficacy of ENDOMETH to replace DL-methionine in broiler diets

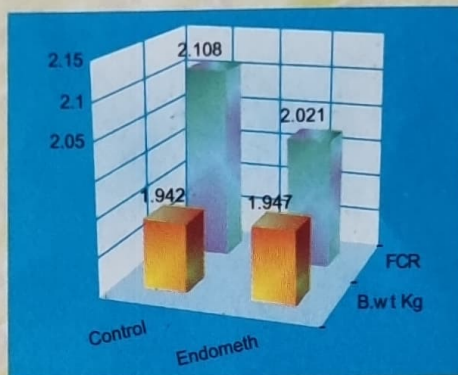
A trial was conducted in commercial broilers to evaluate the efficacy of Endometh to replace DL-methionine in broiler diets and its economical impact.

The experiment consisted of 580 day-old chicks divided into two groups and reared on common starter and finisher broiler diets, supplemented with either DL-methionine (Control) or equivalent proportion of Endometh (Endometh). The birds were reared for 42 days under common conditions during which observations were made.

The birds were observed weekly for body weight and feed intake, and FCR was calculated. At the end of the rearing, 10% of birds from each group were tested for antibody titres against IBD (by ELISA) and ND (by HI). Finally, observations were compiled and the results were compared.

Results pertaining to body weight and FCR are given in fig. 1. **Birds from Endometh group were weighing on an average 5 gms more, and had consumed 153 gms per bird lesser, resulting in a much better FCR (2.02) with out oil.** Immunity as indicated by antibody levels was similar for both the groups. After calculating the Feed cost it was found that **Endometh resulted in an additional returns of Rs. 1260/- per 1000 broilers as compared to DL-methionine.**

The above trial demonstrates the ability of Endometh to bring about a greater reconversion of methionine and hence a better nutrient utilization in poultry, especially in broilers. In the absence of such an influence, it is assumed that the pure DL-methionine activity may only try to compensate the growth requirements by increasing the feed consumption. However, such an increase in feed consumption may not be accompanied by an equivalent growth rate, thus resulting in a poorer FCR.



Note :

1. Farmer Rate of DLM taken as Rs. 185/-per Kg. Farmer Rate of Endometh taken as Rs. 150/- per Kg Difference/Kg-35/-
2. If DLM is totally replaced with Endometh, reduction in feed cost is : Starter-52.5Rs/ton(@1500g/ton additional methionine) and Finisher-35Rs/Kg (@1000g/ton of additional methionine).
3. Broiler farm-gate price is taken as Rs. 30/- for calculations.

The trial was carried out under the Technical Supervision and Guidance of

Dr. B.S.Venkatarami Reddy,
Professor & Head,
Department of Animal Nutrition,
Veterinary College,
Univ. Agricultural Science
Hebbal, Bangalore - 560 024



Manufactured by :

Vesper Pharmaceuticals

Village Singapura, Vidyaranyapura Post,
Bangalore - 560 097 INDIA.

Ph/Fax : +91-80-2838 8942, 2838 8571.